The New Fama Puzzle Lives in Risk-Off Quarters

The forward-premium regression for ten currencies, 1979-2025

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Abstract

Since 2008 the Fama regression of dollar exchange-rate changes on interest differentials has had a positive slope, the new Fama puzzle. In ten currencies through 2025 that positive slope comes from risk-off quarters. In the tenth of quarters with the largest rise in the VIX, the slope moved from −1.89 before 2008 to 6.79 after, with a standard error of 1.89, and it is 4.77 without the 2008 crisis quarter. In all other quarters it is close to zero in both periods. Since 2008, then, high-interest currencies have fallen well beyond their rate differential against the dollar when risk rises, in line with the dollar's role as a safe haven. Over all quarters the post-2008 slope of 1.53, with a standard error of 1.44, cannot be told apart from parity or from the pre-2008 slope of −0.66, because interest differentials have varied half as much. Lagged inflation differentials predict currency excess returns in both periods, and interest differentials stop predicting them after 2008.

1 The regression

Fama (1984) regressed the change in the exchange rate on the forward premium,

$$s_{t+3} - s_t = \alpha + \beta\,(f_t - s_t) + \varepsilon_{t+3},$$

where s is the log dollar price of a foreign currency and the horizon is three months. Uncovered interest parity with rational expectations implies $\beta = 1$, so that a currency with a lower interest rate than the dollar is expected to appreciate by the difference. The classic estimates were negative, which suggests that high-interest currencies tend to appreciate and that carry trades earn the interest differential and more. Bussière, Chinn, Ferrara and Heipertz (2022) show that the slope turned positive during and after the 2008 crisis and call this the new Fama puzzle.

We use end-of-month spot rates from the Federal Reserve's H.10 series and three-month interbank rates from the OECD, both from FRED, for ten currencies against the dollar: the pound, euro, yen, Swiss franc, Canadian, Australian and New Zealand dollars, and the Swedish, Norwegian and Danish kroner. By covered interest parity the forward premium equals the three-month interest differential, $f_t - s_t = (i^{\$}_t - i^{*}_t)/4$, and we use the differential in its place. The interest rates are monthly averages, so the differential is measured with some error relative to the end-of-month spot rate, and covered parity has failed by tens of basis points since 2008 (Du, Tepper and Verdelhan 2018), small relative to the differentials here. The sample starts in 1979, or later where data begin (Swedish krona 1982, Danish krone 1987, euro and Swiss franc 1999); the Japanese rate before 2002 is the three-month certificate of deposit rate. All regressions use non-overlapping quarter-end observations, currency fixed effects unless stated otherwise, and standard errors clustered by quarter.

2 The sign flip and its precision

Table 1 reports the pooled slopes. Before 2008 the slope is −0.66, and −1.27 with a standard error of 0.44 once the 1992-93 crisis in the European exchange-rate mechanism is excluded. After 2008 the point estimate is 1.53, in line with the new Fama puzzle, and currency by currency nine of ten slopes are negative over 1979-2007 and all ten are positive over 2008-2025, although these estimates share a common dollar component. The post-2008 estimate is imprecise, and its 95 percent interval, from about −1.3 to 4.4, contains both parity and the pre-2008 slope. A test of equal slopes before and after 2008 gives $p = 0.16$.

The imprecision comes from the regressor. The within-currency standard deviation of the annualized differential was 2.9 percentage points over 1979-2007, 1.4 over 2008-2025, and 0.5 over 2010-2015, when most policy rates were near zero. Engel, Kazakova, Wang and Xiang (2022) find the same loss of predictive power for the dollar near the zero lower bound. As a result, in rolling ten-year windows (Figure 1) the slope jumps by several units whenever the 1992-93 or the 2008-09 crisis enters or leaves the window.

Sample	Slope β	Standard error	Currency-quarters
1979-2007	−0.66	0.59	953
excluding the 1992-93 ERM crisis	−1.27	0.44	913
2008-2025	1.53	1.44	720
excluding the 2008-09 crisis	0.55	1.24	670
2010-2025	0.53	1.37	640
2016-2025	0.32	1.85	400
1979-2025	−0.35	0.50	1,673
excluding both crises	−0.94	0.40	1,583

Table 1: Pooled Fama regressions of three-month dollar exchange-rate changes on the three-month interest differential. Parity predicts a slope of one. The ERM crisis drops the five quarterly observations dated June 1992 to June 1993, and the 2008-09 crisis those dated June 2008 to June 2009, which cover exchange-rate changes from July 1992 to September 1993 and from July 2008 to September 2009.

	1979-2007	2008-2025
A. Slope of exchange-rate change on differential, by quarter type (from 1986)		
Risk-off quarters (top tenth of VIX increases)	−1.89 (0.96)	6.79 (1.89)
excluding the quarter from June 2008		4.77 (1.60)
Other quarters	−0.16 (0.78)	0.70 (1.39)
B. Slope on components of the differential, no fixed effects		
Cross-currency component (average differential)	0.65 (0.32)	0.45 (0.80)
Cross-time component (average across currencies)	−0.99 (0.68)	1.50 (1.66)
Currency-specific time variation	−0.01 (0.54)	1.60 (1.31)
C. Excess return on foreign deposits, next three months		
On the interest differential	−1.66 (0.59)	0.53 (1.44)
On the lagged inflation differential	−0.24 (0.10)	−0.73 (0.28)
inflation, controlling for the interest differential	0.01 (0.15)	−0.75 (0.28)
D. Cumulative excess return on foreign deposits on today's differential		
Next 4 quarters	−5.74 (1.13)	0.09 (2.59)
Next 20 quarters	−7.75 (2.21)	−4.93 (3.45)

Table 2: Pooled regressions with standard errors clustered by quarter in parentheses. Panel A uses quarters from 1986, when the VXO begins, and classifies a quarter as risk-off if its three-month VIX increase (VXO before 1990) is in the top tenth of all quarters; there are 7 such quarters before 2008 and 9 after. Panel B splits the differential into its currency mean, its cross-currency mean at each date, and the remainder. In Panels C and D the dependent variable is the return on a three-month foreign deposit funded in dollars, $s_{t+3} - s_t - (f_t - s_t)$, for which parity predicts a slope of zero; the inflation differential is US minus foreign year-on-year CPI inflation, lagged one month (one quarter for Australia and New Zealand), with German CPI for the euro, and the CPI sample ends between 2021 and 2025 depending on the country.

3 Four checks suggested by recent work

Risk-off quarters. Since 2007 the dollar has moved with global risk proxies and behaved as a safe-haven currency (Lilley, Maggiori, Neiman and Schreger 2022), and carry trades are exposed to sudden crashes of high-interest currencies (Brunnermeier, Nagel and Pedersen 2008). Panel A of Table 2 splits quarters by the rise in the VIX over the holding period. In the tenth of quarters with the largest increase the slope moved from −1.89 before 2008 to 6.79 after, and dropping the quarter that begins in June 2008 leaves 4.77 with a standard error of 1.60. In the other nine tenths the slope is −0.16 before 2008 and 0.70 after, both indistinguishable from zero. With the top fifth of quarters as the cutoff the risk-off slopes are −1.88 and 6.68. The positive post-2008 slope therefore comes from episodes in which risk rose and high-interest currencies fell against the dollar by much more than their interest differential. The classification uses the VIX change over the same quarter as the exchange-rate change, so it locates the source of the slope after the fact.

Which component of the differential. Hassan and Mano (2019) separate the forward premium puzzle, a time-series phenomenon, from the cross-currency carry trade, and attribute the puzzle to variation over time that is common across currencies. Panel B reproduces this split. Before 2008 the negative slope comes from the cross-time component, −0.99, while the slope on the cross-currency component is positive, 0.65,

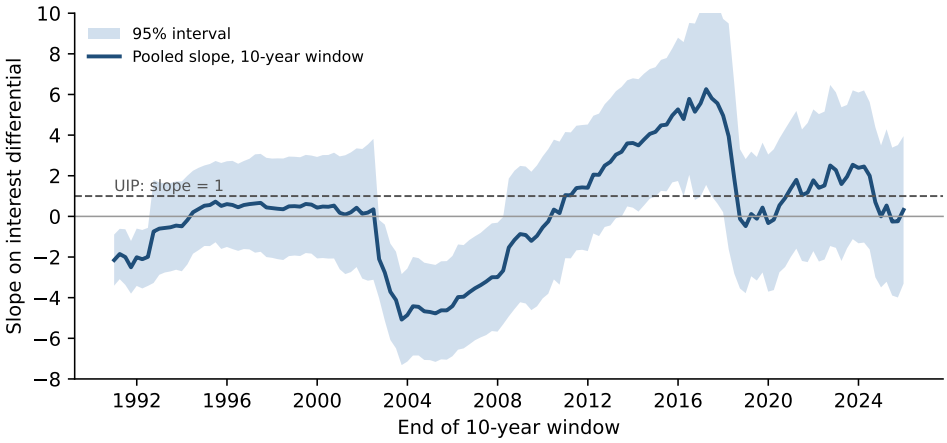


Figure 1: Pooled Fama slope in rolling ten-year windows of quarter-end observations, plotted at the end of each window, with a 95 percent interval. The dashed line is the parity value of one.

so currencies with persistently high rates tended to depreciate, consistent with Hassan and Mano's finding that high-interest currencies need not be expected to appreciate. After 2008 both time components turn positive, 1.50 and 1.60, with standard errors above one.

Inflation differentials as predictors. Engel, Kazakova, Wang and Xiang (2022) find that lagged inflation differentials predict dollar excess returns more consistently than interest differentials. Panel C shows the same pattern through 2025. The slope of foreign excess returns on the interest differential is -1.66 before 2008 and 0.53 with a standard error of 1.44 after. The slope on the lagged inflation differential is negative in both periods, -0.24 and -0.73, so a one percentage point higher US inflation rate predicts a 0.73 percent higher three-month return on dollar deposits after 2008. After 2008 inflation keeps its coefficient when the interest differential is added, and before 2008 the two predictors overlap. Excluding the 2008-09 crisis lowers the post-2008 inflation coefficient to -0.44 with a standard error of 0.24.

Longer horizons. Engel (2016) shows that high-interest currencies are stronger in level than expected real interest differentials imply, and Valchev (2020) models a reversal of the puzzle at horizons of several years. In Panel D the dependent variable is the cumulative excess return on foreign deposits. Before 2008 a foreign rate one percentage point above the dollar rate predicts an excess return of about 1.4 percent over the next year and 1.9 percent over five years, so the quarterly excess returns turn slightly negative after about three years and reverse only a small part of the first-year gain. After 2008 the one-year slope is zero and the five-year slope is imprecise.

4 Scope

These are dollar-based regressions with interest differentials in place of forward rates, three-month horizons and ten advanced-economy currencies; forward rates, other base currencies or emerging-market currencies could give different magnitudes. The crisis windows are dated by hand, and the risk-off split uses contemporaneous volatility. Bussière, Chinn, Ferrara and Heipertz (2022) attribute the post-crisis switch to a change in how expectation errors co-move with interest differentials, using survey expectations that we do not use, and Kremens and Martin (2019) show that risk premia implied by quanto index contracts forecast currency appreciation better than parity out of sample without relying on interest differentials.

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